P3 — The Mathematica notebooks of the project folder: which were used, and how

Provenance required by the task specification (PROJECT spacetimes)

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1. What this document is for

The task specification asks for a provenance that lists every Mathematica notebook that was used and says how it was used. The project folder contains fourteen notebooks. This document lists them, describes the tools with which they were read, and states, cell by cell where it matters, what the project took from each. The *symbols* of the notebooks, and the many places where two notebooks use the same symbol for different objects, are the subject of the MASTER provenance (`provenance/P1_master_symbols`).

The result in short:

- `4+4.nb` is the source of the eight real 16×16 Dirac matrices of the 4+4 spacetime, as the task specification prescribes.
- `7+1.nb` is the source of the eight real 16×16 Dirac matrices of the 7+1 spacetime, as the task specification prescribes, and is quoted in a few places for context.
- The meaning of the eight coordinates is taken from the head of `4+4.nb` and from `Dirac_Einstein-Lovelock_4+4_Nash.nb`.
- The other eleven notebooks were read for their conventions and for the survey of symbols. Nothing is taken from them; most belong to the second stage of the author's programme (gravity), which the task specification says is not to be implemented.
- **Mathematica was not run.** The project reads what Mathematica *printed* in the notebooks, rebuilds every matrix from its definition with its own program (Rust), compares the two entry by entry, and repeats the essential checks with an independent program (Python). Nothing in the project depends on evaluating a notebook again.

The notebooks are not part of the repository (they are the author's unpublished work); `provenance/inputs_sha256.txt` identifies the copies that were read.

2. How the notebooks were read

A notebook file (`.nb`) is a text file that contains one large expression: a list of cells, each holding typeset "boxes". Three small programs, all in the repository, read these files without Mathematica.

Program	What it does	Used for
`tools/nb2txt.py`	converts a notebook to plain text, cell by cell, with a running cell number and the style of each cell (Input, Output, Text, …); typeset input appears in linear form	reading; every citation "cell *n*" in the project refers to its numbering

Program	What it does	Used for
`tools/extract_nb_matrices.py`	reads the integer matrices that Mathematica printed in given output cells	`data/notebook_matrices_4+4.txt`, `data/notebook_matrices_7+1.txt`, against which the programs compare their matrices
`tools/scan_notebook_symbols.py`	lists the symbols that each notebook defines	the survey of symbols of P1

To repeat the reading (from the root of the repository, with the notebooks in that folder):

```
python3 tools/nb2txt.py "4+4.nb" /tmp/4+4.txt 300         # text of a notebook; outputs cut after
 ↪  300 characters
python3 tools/extract_nb_matrices.py "4+4.nb" 13 136 163 165 314 > data/notebook_matrices_4+4.txt
python3 tools/extract_nb_matrices.py "7+1.nb" 204 > data/notebook_matrices_7+1.txt
python3 tools/scan_notebook_symbols.py                     # symbols of all notebooks
```

3. The fourteen notebooks

“Cells” is the number of cells, with the number of input cells in parentheses.

Id	File	Cells
N01	`4+4.nb`	806 (366)
N02	`7+1.nb`	1402 (189)
N03	`Bessel product.nb`	184 (88)
N04	`Dirac_Einstein-Lovelock_4+4_Nash.nb`	403 (112)
N05	`Dirac_Nash_Identity.nb`	400 (192)
N06	`Einstein-Lovelock-4+4-Nash.nb`	343 (82)
N07	`Einstein-Lovelock-7+1-Nash.nb`	342 (82)
N08	`Einstein-Lovelock-Eternal.nb`	316 (64)
N09	`Generalized _Kronecker_Delta_4+4.nb`	218 (57)
N10	`Pair_Creation_of_Universes_WaveFunctionOfUniverse-4+4-Einstein-Lovelock-Nash.nb`	2098 (991)
N11	`WaveFunctionUniverse-4+4-Einstein-Lovelock-Nash.nb`	929 (452)
N12	`WaveFunctionUniverse-4+4-Einstein-Lovelock.nb`	1317 (615)
N13	`det-g.nb`	20 (8)
N14	`field_equations.nb`	271 (69)

What the notebooks contain, and what the project did with them:

Id	What it contains	Use
N01	the real Dirac matrices of 4+4 and their 8×8 and 4×4 building blocks; split octonions; complex structures	**source of the 4+4 matrices** (Sec. 4)
N02	octonions; real Dirac matrices of 7+1; Dirac equation of a 16-component field; in 27 sections also gravity, a brane picture, cosmology	**source of the 7+1 matrices**; quoted for context (Sec. 5)
N03	identities for products of Bessel functions	not used
N04	Einstein–Lovelock field equations on 4+4 with the package xAct; a text on the origin of the work	**meaning of the coordinates** $x0, \ldots, x7$; the lineage quoted in the Introduction of the manuscripts
N05	identities of real 4×4 matrices	read; not used
N06	field equations on 4+4 (xAct)	read; not used

Id	What it contains	Use
N07	field equations on 7+1 (xAct)	read; not used
N08	a solution with inflating and deflating dimensions	read for the picture of Stage 2 that the task specification describes; not used
N09	generalized Kronecker delta, Lovelock tensors	read; not used
N10	a wave function of the universe on 4+4; repeats the construction of N01	read; not used
N11	as N10	read; not used
N12	as N10	read; not used
N13	determinant of a diagonal metric	read; not used
N14	field equations on 4+4 (xAct)	read; not used

"Read" means: the notebook was converted completely to text, its headings and text cells were read, and every cell entered the survey of symbols. Only for N01, N02 and N04 does a statement of the project rest on the content of a notebook; the following sections list these statements.

4. What was taken from `4+4.nb` (N01)

The project rebuilds the matrices from the definitions in the input cells of the notebook and compares the result with the matrices that Mathematica printed.

Object of the notebook	Symbol of the project	Printed in	Compared by check
`s4by4[h]`, `t4by4[h]`, `ζ[A]`, `τ[A]`, `τ` with an overbar, `σ = σ88`	s_h, t_h, τ^A_{1644}, $\bar{\tau}^A_{1644}$, σ_{88}	Out[76], Out[77] (relations)	`44-B1` to `44-B4`
`T16^A[0]` to `T16^A[7]`	γ^A_{1644}	cell 314, Out[211]	`44-N1`
`T16^A[8]`	ω_{1644}	cell 136, Out[99]	`44-N2`
`η4488`	η_{1644}	cell 13, Out[9]	`44-N3`
`P_L`, `P_R`	P^L_{1644}, P^R_{1644}	cells 163 and 165, Out[112], Out[113]; Out[114] (relations)	`44-P3`, `44-P2`
`σ16`	C^-_{1644}	Out[105], Out[106] (relations)	`44-S1`, `44-S4`
`JT4` = `T16^A[4]`	γ^4_{1644} as a complex structure	—	`44-S5`

The meaning of the coordinates ($x0$ hidden space, $x1, x2, x3$ space, $x4$ time, $x5, x6, x7$ additional times) is stated in a text cell at the head of the notebook.

Not taken from N01: its multiplication table of the split octonions, its frames (`E^A_a`), and its search for complex structures. The Dirac *equation* of the field dirac1644, its current, its Lagrangian and everything that follows are not in the notebook; they are derived in "dirac matrices", Part 1.

5. What was taken from `7+1.nb` (N02)

The notebook labels the time by the index 0; the project labels it by 4, as the task specification demands, and relates the two labellings by the exchange swap_{04} of the index values 0 and 4.

Object of the notebook	Symbol of the project	Where in the notebook	Compared by check
`Γ17[0]` to `Γ17[7]` (real set)	—	cell 204, Out[178]; Out[173] (relations)	`71-N1`, `71-N3`
`Γ17up[A]` $= \eta 17^{AA}$ `Γ17[A]`	$\gamma_{1671}^{\mathrm{swap}_{04}(A)}$	its Sec. 5.3	`71-N2`
`τ0[A]` (octonionic 8×8 matrices)	$\tau_{1671}^{\mathrm{swap}_{04}(A)}$	its Sec. 3.1; Out[175] (block form)	`71-B1`, `71-B2`, `71-D6`
the construction of `Γ17` from `T16` and `S44`	relation between the two sets of the project	Out[176]	`71-B3`, `71-B4`
`Γ17[8]` $=$ `ω17`, `χ17`, `PL17`, `PR17`	ω_{1671}, χ_{1671}, $P_{1671}^{L,R}$	its Secs. 4.6 and 7.10	`71-V1`, `71-P1`, `71-P4`
`βD`, `αD[i]`	β_{1671}, α_{1671}^{j}	its Sec. 7.1	`71-D6`
`dirac16complex` and its Dirac equation	ψ_{1671} (dirac1671), Eq. (2.9) of "dirac matrices"	its Sec. 7.2	`71-D1` and following
`S44`	C_{1644}^{+}	its Sec. 4.1	`44-S2`

Statements of the notebook that the project quotes without using them in a calculation:

Statement	Section of `7+1.nb`	Quoted in
With the signs of N01 (space-like $= +1$) no real 16×16 matrices exist for 7+1; with the opposite signs they do	4.4	"dirac matrices", Sec. 2.4 (the project proves the positive part itself, checks `71-R1`, `71-C1`, `71-C3`)
Plane waves, spin sums and quantization of the field were derived symbolically	7.4 to 7.7	"dirac matrices", Sec. 2.8.2 (tested numerically there)
The two coupled 8-component equations	7.11	"dirac matrices", Sec. 2.8.3
A brane picture with the Standard Model on a brane	17, 18	Supplement S2, as an alternative to the hypothesis examined there
A 4-torus and a 4-sphere as extra spaces	25, 26	Supplement S2, for the isometries of possible extra spaces
A mass window for the unpaired field as dark matter	24	Supplement S3, Sec. 4.3, for comparison

Not taken from N02: its complex set `Γ71`, its 32×32 set, its Kohn–Sham theory of a self-coupled field (its Secs. 8, 9, 15, 16), its gravity (Secs. 11 to 14, 20 to 23) and its cosmology. The interaction of this project is a different one: it is mediated by the Vorton-Scalar, which the notebook does not contain.

The notebook says of itself that it was generated by a script and evaluated in Mathematica, and that each of its claims is tested by a cell of the notebook. The project does not rely on those tests: every statement about the matrices that it uses is tested again by its own programs.

6. What was taken from the other notebooks

N04. Its text cells state the meaning of the coordinates in the same words as N01, and describe the origin of the real representations (Dirac's representation of the covering group of $SO(3,2)$, extended by the author to $SO(3,3)$, $SO(4,4)$ and real 16×16 matrices); the Introduction of the manuscripts repeats this in one sentence, with the references that P2 lists. Its field equations are not used.

N06 to N09, N13, N14 (gravity). The task specification describes Stage 2 of the author's programme (three space dimensions inflate, three time dimensions deflate, the hidden dimension curls

up into a ring) and says: be aware of this, but do not implement it. These notebooks are that stage. The project uses the description only where it discusses what Stage 2 might change (Supplement S1, Secs. 6.3 and 8; Supplement S2; the discussion sections of the manuscripts), and takes no equation from them.

N10 to N12. They repeat the matrices of N01 with identical statements (the survey of symbols shows this) and add a wave function of the universe, which is not part of this project.

N03, N05. Unrelated to the task, or concerned with four-dimensional matrices.

7. Checksums

The following lines of `provenance/inputs_sha256.txt` identify the fourteen copies (SHA-256; verify with `shasum -a 256 <file>` on macOS):

```
740023233eaa0cd24ca0e2e7010094671b1d314b698114c08db94050d3e0aed5  4+4.nb
41d252f9852f210c228f45e1ac956d2d7ef26faeb0062a5880280e1aae0effc7  7+1.nb
5e3d97c18050178f74068925c540d6ac5ca0eaa8f9416cc53b96f6a961959b81  Bessel product.nb
cdbb6a2c0b31ffd6a4fd951746a1e1c3ea5748afd371edc5095cea7e34cf79c0
 ↪  Dirac_Einstein-Lovelock_4+4_Nash.nb
66267b2bf43dc711c6228822948db4da8f4b26b1a74d5839c049a35c5314e842  Dirac_Nash_Identity.nb
e749997952af0ba64e347864cab209d140cc361cd059effaadf3d68410f1f96b  Einstein-Lovelock-4+4-Nash.nb
af47a11d9a5d9f56a46a81b0613626ed3b5b7fd13111ed21bd4d3feee763acdc  Einstein-Lovelock-7+1-Nash.nb
a0e01b0101bcad8ee252591065b3999b0efe816b5eb306af2c4e0abd19c888cb  Einstein-Lovelock-Eternal.nb
23bb4e0c70943e766d9b081a3a399ef29664889aad088041329ce2e065b80afb  Generalized
 ↪  _Kronecker_Delta_4+4.nb
5ee5cb2a95146136ee65636a4aef174f303b6c41da130d2c57e4684f9c8ff69f
 ↪  Pair_Creation_of_Universes_WaveFunctionOfUniverse-4+4-Einstein-Lovelock-Nash.nb
d06875b70d0926bd12fbb3d24cc14f2b425be7feaa9956b581319ff54004fa97
 ↪  WaveFunctionUniverse-4+4-Einstein-Lovelock-Nash.nb
3e0ef380cc36198cbc05da72de26a4218c0a64ab0711fa179a1701c77ca0fb8b
 ↪  WaveFunctionUniverse-4+4-Einstein-Lovelock.nb
afca8dcfd94394eb3acb0ba60444968a062ba7c86bbf8a09e6037a6f87c74e9d  det-g.nb
3bd90c1b93aa29c53e1388a4f73f6ec0645227124dbc8ee217c21fdde6d9a3f9  field_equations.nb
```